STA 4001 Stochastic Process, Spring 2024
Midterm (March 24th, 3:00-5:00 PM)

1. (15 points) The number of families Y visiting a theme park in one day follows a Poisson random variable with mean 100. The number of people in each family independently follow the same distribution as given by $P(K=1)=0.1, P(K=2)=0.3, P(K=3)=0.4, P(K=4)=0.2$. Also, the number of people in each family is independent of the number of families Y. Let X be the total number of people that visit the theme park in one day.

(a) Express $E[X|Y]$ as a function of Y. (5 points)

(b) Compute $Var(E[X|Y])$ and $Var(X|Y)$. (10 points)

For those who need a reminder, let's define the Poisson random variable. A random variable Z, taking on one of the values 0, 1, 2, ..., is said to be a Poisson random variable with parameter λ, if for some $\lambda > 0$,

$$P(X=k) = e^{-\lambda}\frac{\lambda^k}{k!}, \quad \text{for } k \geq 0.$$

Solution:
(a). Denote the number of people in family i by K_i. Then the total number of people that visit the park in one day is

$$X = \sum_{i=1}^{Y} K_i.$$

Consider $Y=y$ for some y, then

$$E[X|Y=y] = E[\sum_{i=1}^{Y} K_i|Y=y] = \sum_{i=1}^{y} E[K_i].$$

Since

$$E[K_i] = \sum_{k=1}^{4} kP(K_i=k) = 1\times 0.1 + 2\times 0.3 + 3\times 0.4 + 4\times 0.2 = 2.7,$$

we get

$$E[X|Y=y] = 2.7y,$$

i.e.,

$$E[X|Y] = 2.7Y.$$

(b).

$$\mathrm{Var}(E[X|Y]) = \mathrm{Var}(2.7Y) = 2.7^2\mathrm{Var}(Y) = 7.29\times\lambda = 729.$$

Since K_i's and Y are independent, we get

$$\mathrm{Var}(X|Y=y) = \mathrm{Var}\left(\sum_{i=1}^{Y} K_i|Y=y\right) = \mathrm{Var}\left(\sum_{i=1}^{y} K_i\right) = \sum_{i=1}^{y}\mathrm{Var}(K_i).$$

The variance of K_i is

$$\mathrm{Var}(K_i) = E[K_i^2] - E[K_i]^2 = 1^2 \times 0.1 + 2^2 \times 0.3 + 3^2 \times 0.4 + 4^2 \times 0.2 - 2.7^2 = 0.81.$$

So

$$\mathrm{Var}(X|Y = y) = 0.81y,$$

i.e.,

$$\mathrm{Var}(X|Y) = 0.81Y.$$

P.S. An alternative approach for calculating $\mathrm{Var}(X|Y)$:

$$\begin{aligned}
\mathrm{Var}(X|Y = y) &= E[X^2|Y = y] - (E[X|Y = y])^2 \\
&= E\left[(\sum_{i=1}^{Y} K_i)^2 | Y = y\right] - \left(E\left[\sum_{i=1}^{Y} K_i\right]\right)^2 \\
&= \sum_{i=1}^{y} E[K_i^2] + \sum_{i \neq j} E[K_i]E[K_j] - \left(E\left[\sum_{i=1}^{Y} K_i\right]\right)^2 \\
&= yE[K_i^2] + y(y-1)E[K_i]^2 - y^2 E[K_i]^2 \\
&= y(E[K_i^2] - E[K_i]^2) \\
&= y\mathrm{Var}[K_i].
\end{aligned}$$

2. (25 points) Consider a two-state Markov chain with $s = \{1, 2\}$. The transition matrix is

$$P = \begin{bmatrix} 0.6 & 0.4 \\ 0.4 & 0.6 \end{bmatrix}.$$

(a) (5 points)

(i) Is this Markov chain irreducible?

(ii) Is this Markov chain recurrent?

(iii) Is this Markov chain aperiodic?

(b) Find the stationary probability $\pi = (\pi_1, \pi_2)$. (5 points)

(c) Is this chain reversible? (5 points)

(d) We know that

$$P^3 = \begin{bmatrix} 0.504 & 0.496 \\ 0.496 & 0.504 \end{bmatrix}.$$

Compare $P(X_3 = 1|X_0 = 1)$ and $P(X_3 = 2|X_0 = 1)$. (5 points)

(e) Compute $\lim_{n\to\infty} P(X_n = 2|X_0 = 1)$. (5 points)

Solution: (a).

(i) Yes.

(ii) Yes.

(iii) Yes.

(b). $\pi = (0.5, 0.5)$.
(c). Yes. $\pi_i P_{ij} = \pi_j P_{ji}$.
(d). $P(X_3 = 1|X_0 = 1) = 0.504$, $P(X_3 = 2|X_0 = 1) = 0.496$.
(e). $\lim_{n\to\infty} P(X_n = 2|X_0 = 1) = \pi_2 = 0.5$.

3. (15 points) Assume a random walk with states $\{1, 2, 3, 4\}$ has the transition matrix

$$P = \begin{bmatrix} 0.5 & 0.5 & 0 & 0 \\ 0.2 & 0.3 & 0 & 0.5 \\ 0 & 0 & 0.7 & 0.3 \\ 0 & 0 & 0.3 & 0.7 \end{bmatrix}.$$

(a) Classify the states $\{1, 2, 3, 4\}$ in terms of communication. (5 points)

(b) Identify the transience and recurrence of each state. (5 points)

(c) Let s_{ij} be the expected time periods the Markov chain is at state j, given started at i. Compute s_{12} and s_{21}. (5 points)

Solution:
(a). Classes: $\{1, 2\}$ and $\{3, 4\}$.
(b). $\{1, 2\}$ transient and $\{3, 4\}$ recurrent.
(c). Since the class $\{1, 2\}$ is transient, the matrix P_T is defined as

$$P_T = \begin{bmatrix} 0.5 & 0.5 \\ 0.2 & 0.3 \end{bmatrix}.$$

Therefore the S matrix is calculated by

$$S = (I - P_T)^{-1} = \begin{bmatrix} 2.8 & 2 \\ 0.8 & 2 \end{bmatrix}.$$

4. (15 points) Recall the Gambler's ruin problem, in each game, the gambler wins 1 dollar with chance $p \neq \frac{1}{2}$ and loses 1 dollar with chance $q = 1 - p$. He will stop playing the game either he owns N dollars or he loses everything. He starts with i dollars. The chance he ends up with owning N dollars is denoted by P_i. We know that

$$P_i = \frac{1 - (q/p)^i}{1 - (q/p)^N}, \quad i = 0, 1, 2, \ldots, N.$$

(a) (5 points) From the above conclusion only, prove that Q_i, the chance he loses everything, is

$$Q_i = \frac{1-(p/q)^{N-i}}{1-(p/q)^N}, \quad i = 0, 1, 2, \ldots, N.$$

(b) Let $N = 10, i = 3, p = 0.6$. By using the formula of P_i, compute the chance that his wealth has ever reached 4 dollars before ending the game. (5 points)

(c) Same setting as (b), by using the formula of Q_i, compute the chance that his wealth has ever reached 2 dollars before ending the game. (5 points)

(**Hint:** Please treat the states 4 and 2 in problems (b) and (c) respectively as absorbing states and modify the MC structure to solve the problem.)

Solution:
(a). Method 1: Make transfer $i \to N-i$ and then swap $i \leftrightarrow N-i$ and $q \leftrightarrow p$, then

$$Q_i = \frac{1-(p/q)^{N-i}}{1-(p/q)^N}, \quad i = 0, 1, 2, \ldots, N.$$

Method 2:

$$Q_i = 1 - P_i = \frac{(q/p)^i - (q/p)^N}{1-(q/p)^N} = \frac{(q/p)^N - (q/p)^i}{1-(q/p)^N} \cdot \frac{(p/q)^N}{(p/q)^N} = \frac{1-(p/q)^{N-i}}{1-(p/q)^N}.$$

(b). We take state 4, which means 4 dollars, as an absorbing state. So the result is the winning probability for the gambler's ruin problem with $N = 4$ starting from 3.

$$\text{Probability} = \frac{1-(q/p)^3}{1-(q/p)^4} = \frac{1-(2/3)^3}{1-(2/3)^4}.$$

(c). We take state 2, which means 2 dollars, as an absorbing state. So the new gambler's ruin problem has a winning state 10, losing state 2, where we start from state 3. It is equivalent to a gambler's ruin problem with $N' = 8$, starting from state 1. So the result is the losing probability for the gambler's ruin problem with $N = 8$ starting from 1.

$$\text{Probability} = \frac{1-(p/q)^7}{1-(p/q)^8} = \frac{1-(3/2)^7}{1-(3/2)^8}.$$

5. (30 points) Suppose a company bought a new machine for production. The machine has three possible states: good, so-so and bad. Once the state becomes bad, it will be repaired in one day and back to the state of good at the cost of 100 dollars.

Without any maintenance, the probability that, after one day of use, the machine turns from good to so-so and from so-so to bad are 0.1 and 0.2, respectively. The machine will not turns from good to bad immediately. (The machine will not turn to a better state without repairing.)

To reduce the chance of bad state, the company has an option of buying a maintenance service at daily cost of 5 dollars. Note that the company only pay for the service when the machine is good or so-so, and does not need to pay for it when it's bad. With maintenance, the probability that, after one day of use, the machine turns from good to so-so and from so-so to bad are reduced to 0.08 and 0.1, respectively.

To help the company to make a decision, we need to figure out whether buying the maintenance could reduce the long-run average cost per day. Next, we will model the state of machine with and without maintenance as two Markov chains.

(a) Write down the two transition matrices corresponding to the cases with and without maintenance, respectively. (6 points)

(b) Compute the stationary distributions of the two Markov chains. (14 points)

(c) Compute the long-run average cost that the company needs to pay per day, without and with the maintenance service, using the stationary distributions in part (b). Then, make a decision based on your results. (10 points)

Solution:

(a). Without maintenance:

$$P = \begin{bmatrix} 0.9 & 0.1 & 0 \\ 0 & 0.8 & 0.2 \\ 1 & 0 & 0 \end{bmatrix}.$$

With maintenance:

$$P = \begin{bmatrix} 0.92 & 0.08 & 0 \\ 0 & 0.9 & 0.1 \\ 1 & 0 & 0 \end{bmatrix}.$$

(b). Without maintenance:

$$\pi = (5/8, 5/16, 1/16).$$

With maintenance:

$$\pi = (25/47, 20/47, 2/47).$$

(c). Without maintenance:

$$\text{cost} = 100 \times \pi_3 = 100/16 = 25/4.$$

With maintenance:

$$\text{cost} = 5 \times (\pi_1 + \pi_2) + 100 \times \pi_3 = 100/16 = 425/47 > 25/4.$$

Without maintenance is better.

END